

SIMATS ENGINEERING

CO3 – ASSESSMENT TOOL 1

CASE STUDY – DEEP LEARNING (MLA04)

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Case Study Question 1: CNN for Automated Medical Image Classification

1. Suitable CNN architecture

A suitable architecture is: Input X-ray image → Convolution + ReLU → Max Pooling → Convolution + ReLU → Max Pooling → deeper Convolution blocks → Flatten/Global Average Pooling → Fully Connected layer → Dropout → Softmax output. The input images can first be resized to a common size and normalized. For two classes, the final output can use two softmax neurons (Normal and Disease) or one sigmoid neuron.

2. Motivation and role of CNN components

CNNs are suitable because they learn visual features automatically and preserve spatial relationships in images. Convolutional layers apply filters to detect local patterns. Early filters learn edges and simple textures, while deeper filters learn complex structures and abnormal regions. Pooling reduces spatial dimensions and computational cost while retaining important features. Fully connected/output layers combine the learned features and produce the final class prediction.

3. Parameter sharing

In a convolutional layer, the same filter weights are reused at different locations of the image. Therefore, a filter does not need separate weights for every pixel position. This greatly reduces trainable parameters compared with a fully connected network, lowers memory and computation requirements, and allows the same visual pattern to be detected wherever it occurs.

4. Regularization

Overfitting can be reduced using data augmentation, dropout and batch normalization. Data augmentation creates variations such as small rotations, translations and brightness changes, improving generalization. Dropout randomly disables some neurons during training so the network does not depend too strongly on particular features. Batch normalization stabilizes and speeds training and can provide a regularizing effect. A validation set and early stopping can also be used.

5. AlexNet or ResNet

ResNet would generally be more appropriate for this medical classification task. AlexNet is an older, comparatively shallow architecture, whereas ResNet supports much deeper networks through residual/skip connections. These connections help information and gradients flow through deep networks, reducing the vanishing-gradient problem. ResNet can learn richer hierarchical features and usually provides stronger classification performance, although it requires more computation than a small AlexNet. For a healthcare system, a suitably sized ResNet can provide a good balance between accuracy and computational requirements.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition

1. CNN architecture design

The system can use a CNN backbone followed by an object-recognition/detection output stage. Camera images are resized and normalized before entering the network. Convolutional blocks extract features at increasing levels of abstraction, with activation and pooling/downsampling between blocks. A deeper backbone such as ResNet can then provide features to an output stage that predicts object classes and, when required for detection, object locations. Training should use diverse images covering different lighting, distances, viewing angles and road conditions.

2. Purpose of layers and progressive feature learning

The input layer receives camera images. Convolutional layers apply learnable filters to detect local patterns. Activation functions such as ReLU introduce non-linearity. Pooling/downsampling reduces spatial size and keeps important information, lowering computation. The final fully connected or detection/output layers convert learned features into class and location predictions. Early filters learn edges, corners and simple textures; middle layers learn shapes and object parts; deeper layers learn high-level patterns associated with cars, pedestrians, signs, bicycles and obstacles.

3. Importance of parameter sharing

Convolution uses the same filter weights across different image locations. This reduces the number of parameters and computational requirements compared with a fully connected image model. It also provides translation-related feature detection: the same learned pattern can be recognized whether it appears near the left, center or right side of an image.

4. Overfitting and regularization

Overfitting may occur when the model learns the training images too closely and performs poorly on new road scenes. Suitable methods include data augmentation, dropout, batch normalization, weight regularization, early stopping and using a sufficiently diverse training dataset. Augmentation should include realistic changes in brightness, scale, position and viewpoint so that the model learns robust features.

5. AlexNet vs ResNet

AlexNet has a comparatively shallow architecture and was an important early CNN model. ResNet is much deeper and uses residual/skip connections, which make deep-network training easier and reduce vanishing-gradient difficulties. ResNet can learn more complex hierarchical features and is generally more parameter-efficient for a given level of depth, although its computational cost depends on the chosen ResNet variant. AlexNet is simpler but has less capacity for complex modern vision tasks.

6. Recommended architecture

ResNet is the better choice for autonomous-vehicle vision when accuracy and robust feature learning are priorities. Its residual connections support deeper feature extraction and more reliable training. However, real-time operation requires controlling computational cost. A smaller ResNet variant, optimized inference, appropriate image resolution and hardware acceleration can provide a practical balance between accuracy, latency and resource use. AlexNet may be useful where very low computational cost is the dominant constraint, but ResNet is more suitable for the complex and safety-critical recognition requirements described in the case.

Conclusion: CNNs are effective for both medical image classification and autonomous-vehicle vision because convolution, parameter sharing and hierarchical feature learning allow useful image features to be learned automatically. Regularization is important for generalization, while ResNet is generally the stronger choice when deeper feature learning and classification/recognition performance are required.